GiftLane | Customers Like You

To: Sarah Okafor, Head of E-Commerce **From:** Data & ML Team
Subject: Recommendation pilot and customer groups - findings **Date:** 13 October 2026

Recommendation: retain the best-seller approach for now. Our personalized prototype works and is fast on this dataset, but it did not outperform a simple best-seller list on future purchases. We recommend a further validation round before using it on product pages.

What we built and why

We cleaned two years of retail transactions and used **Truncated SVD** to summarize which products tend to be bought by similar customers. It handles our mostly empty customer-product table directly and was faster than the PCA alternative in our tests. The 100-component version gave our strongest exploratory SVD recommendation result.

Measure	Pilot result
Purchase-history table	5,474 customers × 3,688 products; 97.95% empty
Storage: dense vs sparse	77.01 MiB vs 3.18 MiB (95.87% less with sparse storage)
100-component factors	Approx. 3.50 MiB for customer + product factors (float32)
Recommendation accuracy: Hit Rate@10	SVD 25.88% | PCA 24.83% | best sellers 26.87%
Time on test machine	SVD fitting 0.31 s; scoring/evaluation 3.99 s

What the results mean

Among **1,913** evaluable customers, SVD correctly included a new purchased item in the top 10 for **495** customers, versus **514** for best sellers. The baseline leads by **0.99 percentage points**. Compressing the data loses fine-grained purchasing information: relative reconstruction error was **0.7808** at 100 components. The 95.87% storage reduction above comes from sparse storage, not SVD compression.

A marketing-ready customer map

We identified five preliminary groups: **Everyday Gifts & Baking** (1,601), **Bags & Accessories** (947), **Regency Tea & Tableware** (442), **Home Décor & Heart Gifts** (1,337), and **Creative Gifts & Activities** (1,147). These suggest themed campaigns and product bundles. The UK is the leading country in every group.

Risks and next steps

New products and customers without purchase histories are difficult to recommend to; the evaluation excluded purchases of products absent from the training catalogue. Segment boundaries also overlap in the two-dimensional map. Model choice was explored using the held-out test period, so we need a separate validation period and fresh final test. **Next:** test a popularity-plus-personalization hybrid, assess purchase recency and frequency, then run a controlled online pilot only if it beats best sellers. No customer IDs appear in this memo.

Figures are from the project's local experiments on UCI Online Retail II (Chen, 2012); run times are not a full production nightly-job benchmark.